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5G defense network using commercial gNodeB with zero trust architecture

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Abstract

As 5G technology expands globally, the demand for secure and reliable military communication is growing significantly. This study examines 5G military network architectures that integrate commercial 5G elements, ranging from isolated to shared configurations, to balance security, cost, and coverage. We propose a network leveraging commercial next-generation Node B (gNodeB or gNB) with a military Public Land Mobile Network (PLMN) ID, ensuring extensive coverage, robust security, and cost efficiency. To enhance secure access, a zero trust (ZT) architecture using a software-defined perimeter (SDP) is employed. The connection process for user equipment accessing private networks is detailed through 5G authentication and ZT access management. Simulated 5G analysis compares scenarios before and after applying SDP under denial-of-service (DoS) and IP scanning attacks. Results demonstrate that SDP mitigates IP scanning threats, improves throughput during DoS attacks, and maintains performance. Validation on a real-world 5G testbed confirms feasibility, robust security, and stability. These findings underscore the potential of ZT-based 5G military networks for modern defense communication.

KEYWORDS

5G military network, cyber security, software-defined perimeter (SDP), trust computing, zero trust (ZT)

1 | INTRODUCTION

Given that 5G technology is being adopted as the next-generation technology for communication, the proliferation of 5G networks is increasing across a growing number of countries worldwide. The release of 5G-compatible devices from multiple manufacturers has promoted the consumer adoption of 5G, leading to a year-over-year global surge in 5G service users [1]. In areas such as the military, the industrial Internet of Things, autonomous vehicles, and telemedicine, the growing demand for higher data speeds, lower latency, and improved connectivity for new applications is accelerating the adoption of 5G technologies. Specifically, there is a growing demand for using 5G networks to access private networks that require high security in defense operation contexts [2–5].

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The United States Department of Defense has devised a strategic plan for implementing 5G mobile networks to discover new applications and encourage technology adoption [6, 7].

Unlike the 5G network architecture for commercial services, 5G networks for military services require greater security and reliability. From the perspective of 5G network architecture, the most secure method to provide military 5G communication services is to establish a private 5G network architecture specifically for military groups [8]. However, establishing a full-stack private 5G network exclusively for military services has drawbacks such as high initial deployment and maintenance costs, limited coverage, and the need to use designated frequencies [9]. To address these issues, leveraging commercial 5G network components can lower the initial setup costs and more easily provide broader coverage. However, the use of commercial 5G network components to access defense networks presents significant security concerns [10]. Therefore, we propose a secure 5G architecture for private network access that leverages appropriate commercial 5G network components and employs secure technology.

Typically, virtual private network (VPN) technology is used to access private networks from public networks because VPNs encrypt internet traffic. 5G mobile devices in public areas can also use VPN to access private networks [11, 12]. However, accessing private networks using VPN applications and servers does not utilize the features of the 5G architecture, which involves diverse core network functions. Furthermore, traditional VPN technology, which performs authentication only after allowing access based on static perimeter security, contradicts the zero-trust (ZT) principle, which assumes that nothing can be trusted by default. To improve and complement the limitations of naive VPN services, the ZT model—which emphasizes "never trust, always verify"—has emerged [13]. The United States Department of Defense also introduced ZT as a necessary paradigm shift in cybersecurity, moving from classical perimeter-based security to a data-centric security model [14].

Active efforts have been made to apply ZT to 5G mobile networks. In [15], a sustainable ZT security framework for 5G mobile core networks was proposed. The authors provided security within the mobile core network using a ZT model that combined a virtualized evolved packet core and virtualized software-defined perimeter (SDP). In [16], a ZT security model was applied to address the trust challenges in network slicing within a dynamic 5G open architecture. The authors introduced VEAbility, a trust metric that evaluates vulnerability, exploitability, and attackability, to monitor the real-time security state of network slices. In addition, user experience indicators were incorporated to enhance trust quantification in the 5G network architecture. Although ZT-based 5G networks enhance security, they often overlook the integration of a comprehensive ZT architecture in scenarios where 5G mobile devices access private or closed networks.

We propose a 5G defense network using a commercial next-generation Node B (gNB) with a ZT architecture. Mobile devices based on the 3rd Generation Partnership Project (3GPP) 5G standard access a defense-specific core network (CN) through commercial gNB using a defense public land mobile network (PLMN) identifier (ID). We consider a situation in which an unregistered general user does not have access to the defense-specific CN in a private defense network. We configure a ZT-based 5G mobile network architecture using an SDP, enabling 5G mobile devices to securely and reliably access private networks. To validate our architecture, we conducted a simulated 5G network analysis using Open 5GS [17] and UERANSIM [18]. This analysis compared performance scenarios before and after applying SDP under denial-of-service (DoS) and IP scanning attacks. The analysis demonstrated that before applying SDP, IP scanning attacks exposed vulnerable network endpoints, rendering the system susceptible to unauthorized access attempts. In a separate DoS attack scenario, the throughput without SDP decreased drastically, reaching approximately 50% of normal capacity. By contrast, when SDP was applied, the network maintained throughput levels above 65%, ensuring stable and reliable performance even during active attacks. Additionally, we measured the connection setup time and found that while SDP added a slight latency, the tradeoff significantly enhanced network security and resilience. We implemented a real-world 5G ZT testbed using an open-air interface (OAI) open-source project [19]. This testbed included 5G mobile devices with a Quectel RM500Q-GL, software-defined radio (SDR) USRP [20] gNB, and a 3GPP standard-based CN. In this testbed, we experimentally verified the ZT functions and user scenarios of the proposed architecture using a 5G commercial mobile terminal to access a private network. The primary contributions of this study are as follows:

- 5G defense network architecture leveraging commercial gNB is described, and its security, cost, and coverage are compared with other private 5G architecture scenarios.
- A ZT-based 5G defense network architecture utilizing SDP was proposed and implemented.
- The performance of the proposed ZT-based 5G network was analyzed and evaluated via simulations, demonstrating its resilience to DoS and IP scanning attacks.
- The proposed architecture is deployed and tested in a real-world experimental environment using commercial equipment to verify its feasibility and effectiveness.

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The paper is organized as follows: Section 2 provides an overview of the foundational information. Section 3 describes 5G defense network architectures that leverage commercial 5G network components to access private defense networks. Section 4 details the proposed ZT-based 5G defense network architecture and communication flow. Section 5 presents simulation results evaluating the performance of the ZT-based 5G defense network. Section 6 verifies 5G use cases for the proposed architecture on a real-world testbed. Section 7 discusses the implications of the results and future research directions. Finally, Section 8 concludes the paper by summarizing the key contributions.

2 | PRELIMINARY INFORMATION

2.1 | 5G network architecture

The 5G network architecture was structured to accommodate the following essential elements, with key contents and communication processes outlined as follows [21]:

- User equipment (UE): Refers to wireless devices connected to 5G networks, including smartphones, tablets, laptops, and IoT devices.
- gNB: A 5G base station that facilitates wireless communication between user devices and the 5G core network, offering higher data speeds, reduced latency, and greater capacity than the 4G eNodeB.
- CN: The central component of a 5G system, responsible for handling data traffic, user connections, and network functions. It uses network slicing and edge computing to improve performance and flexibility.
- Control plane (CP): Manages signaling processes such as user authentication, session management, mobility management, and network policy enforcement to control data flow and user access.
- User plane (UP): Handles the transmission of user data, ensuring quality of service (QoS) and managing traffic between the network and user devices.
- Non-standalone (NSA)/Standalone (SA) [22]: NSA mode uses existing 4G LTE core functions to deliver 5G services and is primarily adopted in early 5G deployments. In contrast, SA mode operates independently of 4G LTE infrastructure, offering lower latency and higher data speeds.
- PLMN ID: A PLMN ID consists of a mobile country code and a mobile network code. It identifies a mobile operator within a country and connects user devices to the corresponding CN.

In a 5G network using UE, a gNB, and an SA CN, the connection process involves interactions across both the CP and UP as follows: When powered on, the UE scans available frequency bands to detect nearby gNBs and evaluate their signal strength. It selects a PLMN ID based on scan results and connects to its mobile operator. The gNB forwards the UE's signal to the CN using the selected PLMN ID. In the CP, the CN authenticates the UE using its international mobile subscriber identity (IMSI) and the authentication key stored in the universal subscriber identity module (USIM). Upon successful authentication, a security-key exchange is performed to enable encrypted communication. Unlike earlier generations, 5G mandates end-to-end encryption and protects the IMSI by using a subscription concealed identifier (SUCI)—a temporary encrypted identifier generated on the UE to prevent plain-text IMSI transmission, enhancing protection against identity tracking and eavesdropping. After authentication, the CN establishes a session with the user plane function (UPF) and assigns an IP address to the UE for data communication. The UP then manages user data transmission between the UE and the network, ensuring QoS as defined by the CP. This process enables secure and efficient access to 5G services, including internet connectivity, voice calls, messaging, and streaming.

This 5G architecture ensures secure and efficient communication, with the CP managing signaling and control and the UP handling user data transmission. Additionally, the modular design of the 5G core allows for secure isolation of network slices, reducing cross-slice threats and enhancing security compared to previous generations of mobile communication [23].

2.2 | SDP for ZT architecture

The ZT architecture is a security model that does not distinguish between internal and external networks; instead, it requires all access to be authenticated and continuously verified. This model enables organizations to enforce the principle of least privilege and protect data from both internal and external threats. An SDP is a method for establishing virtual networks protected through authentication and access control, and it represents a practical implementation of ZT [24]. Unlike traditional VPN technology, which authenticates after a network connection is established, SDP enhances security by enforcing access authorization along with bidirectional encryption. SDP also ensures secure communication by separating the control channel—used for authentication and access control—from the data channel used for actual data

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transmission. These channels are protected by secure tunnels, such as IPsec or transport layer security (TLS), to maintain confidentiality and integrity. Additionally, multifactor authentication methods, such as one-time passwords or public key infrastructure, are employed through the control channel to further strengthen security. The SDP approach significantly enhances the security of private networks that utilize VPNs. In [25], the SDP consists of two components: controllers and hosts. The SDP hosts include initiating (IH) and accepting (AH) hosts. In Figure 1, the SDP controller, which plays a central role in all authentication and access controls, comes online, and the system becomes ready (item 1). The SDP controller initiates authentication to verify the identities of the AHs and IH (items 2 and 3). Once authentication is successful, secure control channels are established between the hosts and SDP controller. After the IH and AHs are authenticated and authorized, the SDP controller provides the AHs with a list of authorized IHs (items 4 and 5). The IH then receives the IP addresses of the AHs from the SDP controller (item 6). Using this information, the IH establishes secure data channels with the AHs to enable encrypted communication (item 7). These data channels ensure secure data transmission using protocols such as Internet Protocol Security (IPSec) and TLS. The SDP controller continuously monitors network activity in real time and records logs. It also ensures compliance with security policies by regularly auditing host activity.

3 | 5G DEFENSE NETWORK ARCHITECTURE SCENARIO

To access a private network via 5G communication, various 5G network elements, such as gNB and CN functions, are utilized. For private defense networks with high-security requirements, building a private full-stack 5G network architecture that operates all network elements independently of the commercial infrastructure is a secure option. However, constructing and maintaining a complete set of 5G elements solely for defense comes with high costs [26]. Figure 2 illustrates four distinct 5G network architecture scenarios that integrate both commercial and defense-specific 5G elements to connect to a closed defense network. These scenarios highlight the trade-offs between using commercial elements and ensuring defense-grade security. Each scenario differs in its balance of security, cost, and reliance on commercial components.

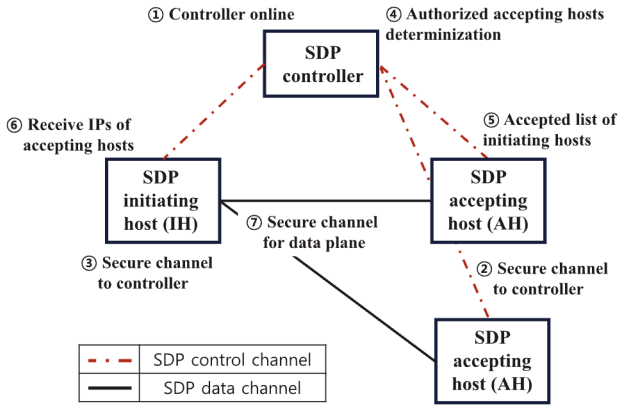


FIGURE 1 Connection process of the software-defined perimeter (SDP) for establishing a secure channel.

- Scenario 2.A: Fully isolated defense network using exclusively defense-specific gNB, CP, and UP elements.
- Scenario 2.B: Shared gNB and CN CPs with a defense-specific UP for enhanced security.
- Scenario 2.C: Network slicing approach, where commercial and defense traffic coexist within the same infrastructure but remain logically separated.
- Scenario 2.D: Commercial gNB integrated with a defense-specific CN (PLMN ID-based access) offering cost efficiency and broader coverage while maintaining security.

Figure 2A illustrates a 5G defense network architecture that is fully isolated from a commercial 5G network. In this architecture, the defense-dedicated gNB operates on defense-specific frequency bands, ensuring complete separation from commercial networks. This isolation provides enhanced security by protecting the defense network from external cyberthreats. However, the high cost of constructing and maintaining an independent network makes it difficult to achieve widespread coverage. Consequently, this approach is most suitable for small or specific areas where high security is critical [27, 28].

Figure 2B depicts a scenario in which a military-dedicated CN UP is implemented while sharing a commercial gNB and CN CP. In this architecture, the commercial CN CP manages the CP services, such as 5G authentication and handover, whereas the military CN UP handles data traffic. This approach lowers costs by leveraging commercial gNB and CN CP infrastructure. However, it introduces security risks because sensitive military information, including subscriber privacy and location data, could be exposed to the commercial CN CP. This architecture is best suited for defense communications requiring broad coverage and moderate security.

Figure 2C shows a 5G military network architecture based on network slicing [29, 30]. In this scenario, all 5G

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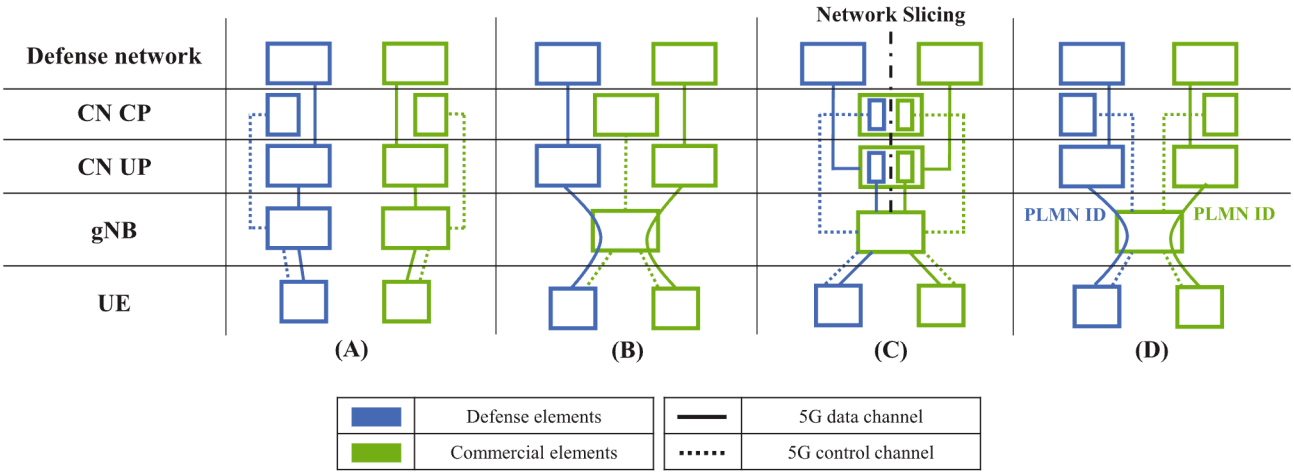


FIGURE 2 Four 5G defense network architecture scenarios for connecting a closed private network using commercial 5G components. (A) Fully isolated defense network separated from commercial 5G infrastructure; (B) 5G defense network architecture of commercial gNB and control plane (CP) of the core network (CN) using the defense user plane (UP); (C) integration of 5G slicing techniques with commercial 5G components for defense network deployment; and (D) deployment using commercial gNB and military CN with military public land mobile network (PLMN) identifiers (ID). The blue components represent commercial network elements typically shared with public users, whereas the green components indicate defense-specific elements dedicated to securing military communication.

elements except the UE and gNB are shared with commercial components, with a military-specific slice providing logical separation. However, because both the CN CP and CN UP interact with commercial infrastructure, the risk of unauthorized access or data leakage increases, particularly without strong encryption and strict access control. Moreover, network slicing technology and service standardization are still under development, making operational deployment challenging. Despite these limitations, network slicing offers low construction costs and operational flexibility, making it a promising option for future battlefield applications using advanced 5G technologies [27].

Figure 2D presents a 5G network architecture that uses a commercial gNB while employing military-owned CN UP and CN CP within the military data network. This scenario leverages the PLMN ID to distinguish military from commercial traffic and ensure secure, efficient routing. It allows the use of the same frequency bands as commercial UEs and eliminates the need for constructing a separate gNB, significantly reducing initial deployment costs while providing broad coverage. Furthermore, because CP and UP are owned and managed by the military, high security is maintained. However, commercial gNBs must support the military PLMN ID to correctly route traffic to the military CN, requiring coordination and infrastructure updates from commercial operators. This architecture enables the delivery of various military services that demand extensive coverage and high security at a relatively practical cost.

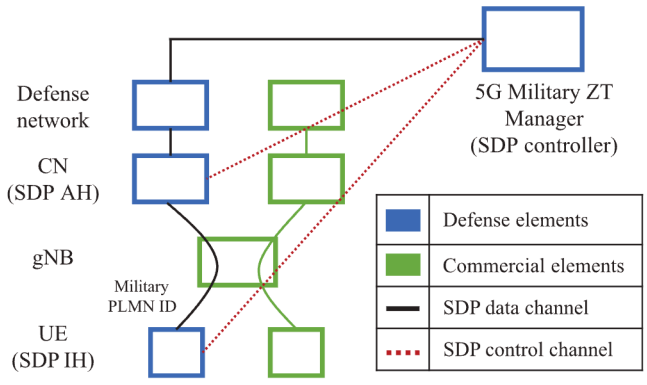


FIGURE 3 Conceptual diagram of a zero-trust (ZT)-based 5G defense network architecture leveraging commercial gNodeBs.

4 | ZERO-TRUST 5G DEFENSE NETWORK ARCHITECTURE

This section presents the proposed ZT-based 5G defense network architecture, detailing the process by which the UE connects to the defense network using 5G standard authentication and ZT-based secure connectivity. As shown in Figure 3, our network uses a commercial gNB integrated with a ZT architecture. The defense network is a private network accessible only to military personnel, using a military-specific PLMN ID to connect the military-dedicated CN (CP and UP) via a commercial gNB. To manage the components of the 5G defense network based on ZT, we applied an SDP architecture consisting of SDP

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IH, AH, and a controller. Military UEs based on commercial frequency bands interact with SDP IHs, while application servers in the private defense network that serve as destinations for the UEs also function as SDP IHs. The military CN acts as the SDP AH, accepting connection requests from authenticated IHs and facilitating communications between them. The SDP control channels (red dashed lines) manage authentication and access control, whereas the data channels (black solid lines) enable secure communication between components.

In private data networks, a 5G military ZT manager functions as the SDP controller. Located within the private network, it blocks access from commercial UEs, enabling secure and efficient management by military administrators. The ZT manager stores and manages all the authorized devices and subscribers with access to the military network via 5G communication. After standard 5G authentication, the UE (as SDP IH) and the CN (as SDP AH) initiate SDP-based authentication through the SDP controller. The UE communicates with the application servers in the data network via the gNB and UPF. The SDP control channel manages authentication and policy enforcement, ensuring secure access to the network, whereas the SDP data channel enables encrypted data transmission between authenticated components (Figure 3). Both channels are encrypted using protocols such as IPSec or TLS to ensure confidentiality and integrity. Additionally, the ZT manager exchanges SDP control messages with SDP IHs and AHs over the control channel to maintain secure, authenticated communication across the network.

4.1 | Communication flow

The network architecture shown in Figure 4 includes a 5G communication network and a defense network hosting a military application server. The 5G CN functions as a military-specific core, accessible only via a dedicated PLMN ID (Figure 3). This communication flow outlines how a 5G UE securely connects to a server within the defense network. The 5G data channel is encrypted in accordance with 5G standards, while both the SDP control and data channels use IPSec or TLS to ensure confidentiality and integrity.

Communication begins with the preparation of the 5G ZT network. At this stage, the military ZT manager, acting as the SDP controller, pre-registers and provisions 5G military subscribers and devices. This involves entering the information necessary for SDP authentication and secure channel establishment. The ZT manager sets up secure control channels with both the CN (as SDP

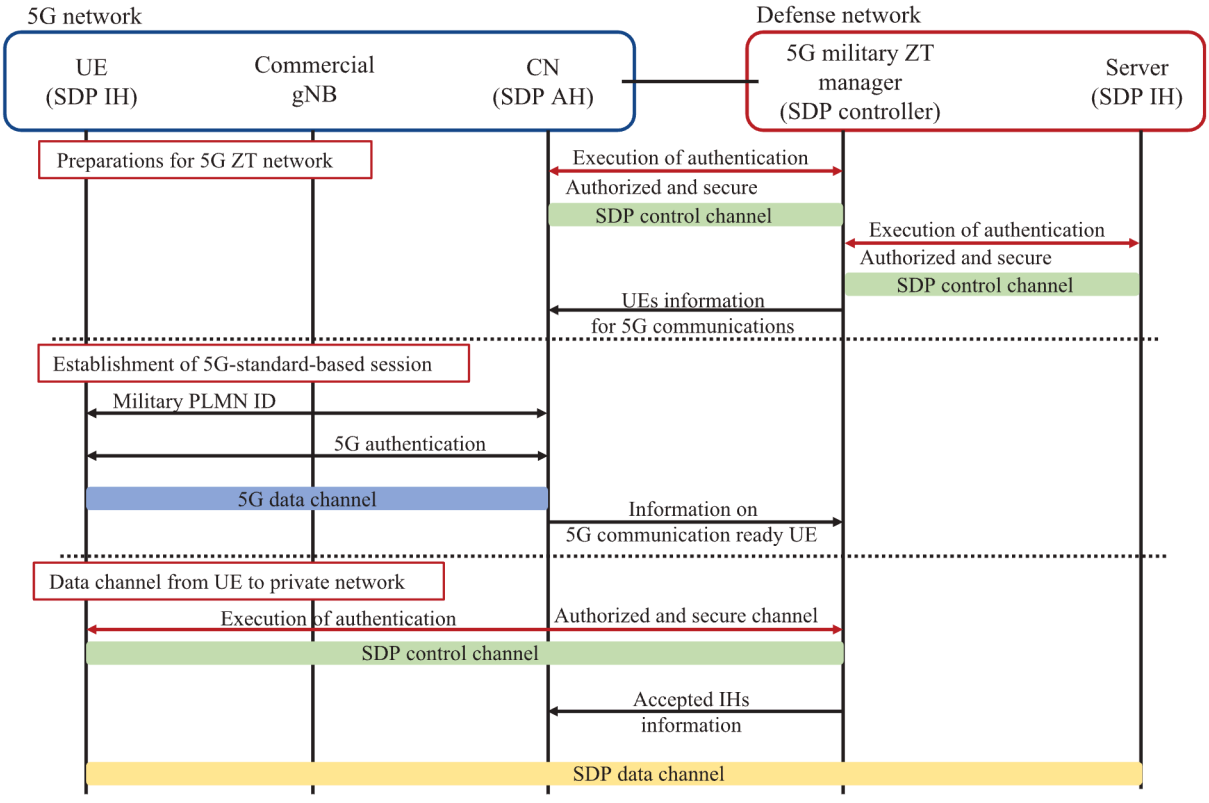


FIGURE 4 Communication flow in the proposed defense network using a commercial gNB with SDP components.

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AH) and application server (as SDP IH) to manage authentication and enforce communication policies. It also shares UE information required for 5G authentication with the CN via the SDP control channel, including the credentials and identifiers needed to establish secure 5G-based connections.

Once preparation is complete, the 5G UE initiates a connection with the military CN via a commercial gNB using a dedicated PLMN ID. The military CN authenticates the UE using information provided by the ZT manager. Upon successful authentication, a protocol data unit (PDU) session is established, creating a secure 5G data channel between the UE and CN. The ZT manager then records key network information of the UE, such as its IMSI and IP address. After the data channel is established, the CN enforces default network access restrictions, applying policy and charging control rules [31] or using the UPF's firewall to ensure secure and controlled communication. After the 5G data channel is established, a ZT-based secure communication process is initiated. The UE requests additional authentication from the SDP controller, which verifies its identity and relays the results to the CN. If the UE passes SDP authentication, the ZT manager provides the CN with a list of authenticated IHs and supplies the UE with information about other authenticated IHs. This enables the UE to establish secure communication with the application server within the defense network. If the UE fails authentication, the ZT manager instructs the CN to block the UE from accessing the private network. Finally, an encrypted SDP data channel is established between the UE and the application server, ensuring secure and reliable communication within the defense network.

5 | SIMULATION RESULTS

This section presents the simulation results of the proposed ZT-based 5G defense network architecture. The simulation environment is shown in Figure 5 and detailed in Table 1, which outlines the roles and configurations of the virtual machines (VMs) used in the testbed. The proposed network was evaluated through simulations to assess its performance, demonstrating resilience against DoS and IP scanning attacks.

5.1 | Simulation setup and scenario

Figure 5 illustrates the virtual 5G network environment constructed using the open-source software Open5GS [17] and UERANSIM [18]. The environment was designed to implement the proposed architecture and evaluate its performance. The SDP IH program was implemented in the UE using C++, the SDP AH software in the CN using Python, and the military ZT manager using C++. For the web graphical user interface (GUI) of the ZT manager, the Spring Framework with Java was used. As shown in Table 1, experiments were conducted using six VMs running Ubuntu 22.04, with the network configured

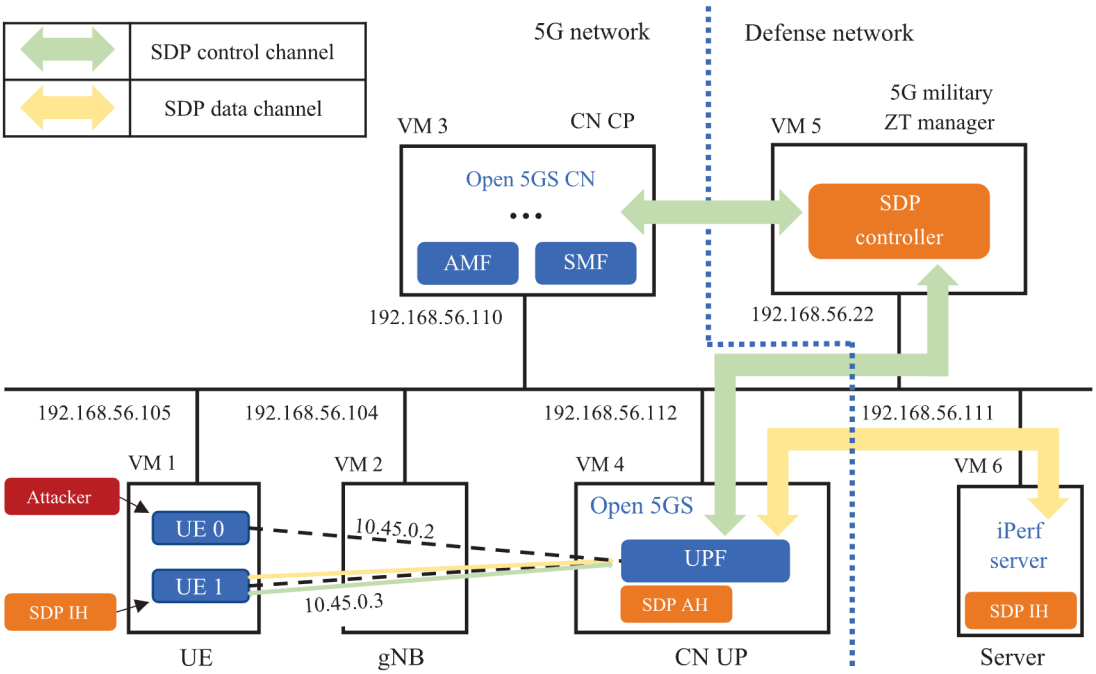


FIGURE 5 Simulation setup architecture of the proposed ZT 5G defense network.

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TABLE 1 Configuration details and assigned roles in the simulation architecture.

VM #	SW and role	IP	Network	SDP role	OS
VM 1	UERANSIM UE	192.168.56.105	5G	SDP IH	Ubuntu 22.04
VM 2	UERANSIM RAN (gNB)	192.168.56.104		-	
VM 3	Open5GS CN CP	192.168.56.110		SDP AH	
VM 4	Open5GS CN UP	192.168.56.112			
VM 5	5G military ZT manager	192.168.56.22	Defense	SDP controller	
VM 6	Server	192.168.56.111		SDP IH	

TABLE 2 Configuration and roles of user equipment (UE).

UE #	IMSI	IP (interface)	Role
UE 0	001010000000000	10.45.0.2 (uesimtun0)	Attacker
UE 1	001010000000001	10.45.0.3 (uesimtun1)	Client

within the 192.168.56.x subnet. The architecture is divided into two main parts: the 5G network and a defense network, with each component playing a specific role.

The 5G network comprises a UE, gNB, and CN (CP and UP). As shown in Table 1, VM 1 runs the UERANSIM UE and simulates two UEs (UE 0 and UE 1). Table 2 shows configuration and roles of UEs. UE 0 is configured as an attacker with IP address 10.45.0.2 (uesimtun0 interface), attempting unauthorized access to the defense network. UE 1, acting as a client, is assigned the IP address 10.45.0.3 (uesimtun1 interface) and accesses the defense network through SDP authentication, serving as an SDP IH. VM 2 runs the UERANSIM gNB, processes connection requests from the UEs, and forwards them to an Open5GS-based CN. VM 3 implements the Open5GS CN CP, which includes key control functions such as access and mobility management and session management. VM 4 runs the Open5GS CN UP, which processes user data traffic and acts as an SDP AH, providing firewall and policy enforcement capabilities.

The defense network consists of a 5G military ZT manager and an application server. VM 5, serving as the core of the ZT architecture, acts as the SDP controller, managing all SDP authentication requests and handling network entities through secure control channels. VM 6 functions as the application server, acting as an SDP IH, allowing data communication only with authenticated UEs. Additionally, an iPerf server was installed on VM 6 to support network performance testing.

In this simulation environment, we measured and compared the time required for a 5G terminal to establish a data communication connection with and without the application of SDP. The simulation results represent the average values obtained from 10 iterations.

TABLE 3 Connection time and average throughput performance.

Criterion		Without SDP	With SDP
Connection time	[32]	0.09 s	1.02 s
	Ours	0.297 s	0.786 s
Average throughput (Without attack)	Ours	299.7 Mbits/s	197.6 Mbits/s
Average throughput (With attack)	[32]	6.86 Mbits/s	1.51 Gbits/s
	Ours	145.9 Mbits/s	194.3 Mbits/s

Additionally, we compared the connection time results with those reported in [32], which evaluated SDP performance in a virtual wired network environment. We also analyzed and compared the throughput performance before and after SDP authentication in scenarios involving DoS or IP scanning attacks. In our simulation, the DoS attack scenario involved a single attacker node (UE 0) repeatedly generating high volumes of TCP traffic, targeting the server to simulate a DoS condition. In the ZT architecture, if the authenticated SDP IH does not send a keep-alive message within 3 s, the SDP data-channel session is terminated to ensure security.

5.2 | Simulation results analysis

Table 3 presents the results of the connection time and average throughput performance in the proposed ZT 5G network architecture under various scenarios, comparing conditions with and without the application of SDP. These results, drawn from our simulation and the findings in [32], provide insights into the impact of SDP on network performance, particularly in terms of connection time and throughput, both under normal conditions and during DoS attacks. In contrast, [32] examines a more severely distributed DoS (DDoS) attack scenario, in which multiple attacker nodes collectively disrupt network operations, providing a broader evaluation of network resilience under stronger attack conditions.

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For connection time, the results indicate that without SDP, our simulation recorded a connection time of 0.297 s, compared to 0.09 s reported in [32]. This difference can be attributed to different network environments: [32] evaluated a virtual wired network, whereas our simulation involved a wireless 5G network. When SDP was applied, connection times increased, as expected, owing to the added authentication processes. [32] reported a connection time of 1.02 s, whereas our simulation achieved a shorter time of 0.786 s. This difference reflects variations in exchanged authentication information and specific procedures used to establish the SDP connection. These results demonstrate that, although SDP introduces some overhead, it remains well-suited for a 5G wireless network environment.

In terms of average throughput performance without attacks, the results show that our simulation achieved 299.7 Mbits/s without SDP and 197.6 Mbits/s. The reduction in throughput when using SDP can be attributed to the overhead introduced by the SDP control and data channels, which provide enhanced security at the cost of minor performance degradation. This decrease is primarily caused by the configuration of our SDP system, where the UE sends keep-alive messages to the SDP controller every second via the SDP control channel to maintain the SDP data channel. Under attack conditions, the impact of the SDP becomes more pronounced. In our simulation, throughput without SDP dropped significantly to 145.9 Mbits/s owing to the effects of a DoS attack. However, when SDP was applied, the throughput increased to 194.3 Mbits/s, indicating SDP's ability to mitigate the adverse effects of such attacks. In comparison, the results from [32] show a much more dramatic impact, with throughput dropping from 1.51 Gbits/s to 6.86 Mbits/s under DDoS attack conditions. The performance in [32] reflects its focus on extreme DDoS scenarios with multiple attacker nodes, demonstrating SDP's ability to maintain communication under severe network disruptions, unlike in our single-node DoS scenario.

Figure 6 shows the network throughput over a 30-second period under four different scenarios: (i) without SDP and without attacks, (ii) with SDP and without attacks, (3) without SDP under a DoS attack, and (4) with SDP under a DoS attack. The results provide valuable insight into the impact of SDP on network performance under both normal and adverse conditions.

In the first scenario, where neither SDP nor DoS attacks were present, the network achieved a stable throughput of approximately 300 Mbits/s, representing the baseline performance. In the second scenario, where SDP was applied in the absence of an attack, the throughput decreased slightly to an average of 197.6 Mbits/s. This

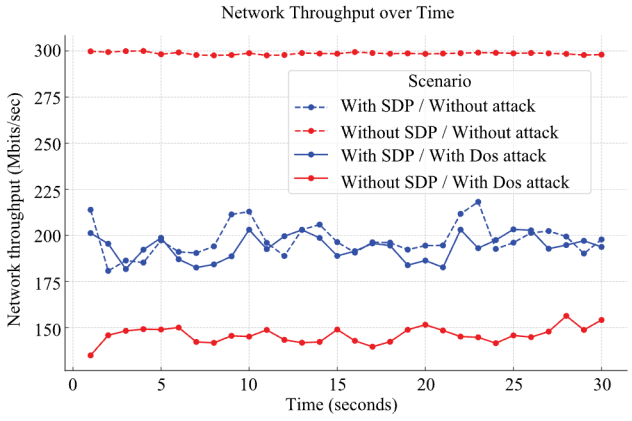


FIGURE 6 Comparative network throughput under SDP and denial-of-service (DOS) attack conditions.

```
root@VM5:/home/vm5# nmap -e uesimtun0 -sn 192.168.56.0/24
Starting Nmap 7.80 ( https://nmap.org )
Nmap scan report for 192.168.56.22
Host is up (0.0037s latency).
Nmap scan report for 192.168.56.100
Host is up (0.0031s latency).
Nmap scan report for 192.168.56.104
Host is up (0.0032s latency).
Nmap scan report for 192.168.56.105
Host is up (0.0032s latency).
Nmap scan report for 192.168.56.110
Host is up (0.0045s latency).
Nmap scan report for 192.168.56.111
Host is up (0.0046s latency).
Nmap scan report for 192.168.56.112
Host is up (0.0010s latency).
Nmap done: 256 IP addresses (7 hosts up) scanned in 66.66 seconds
```

FIGURE 7 IP scanning attacks resulting in the absence of SDP.

reduction is caused by the overhead associated with maintaining the SDP control and data channels. The impact of DoS attacks becomes apparent in the third scenario, where the network operates without an SDP under attack conditions. Here, the throughput decreased to an average of 145.9 Mbits/s, demonstrating the network's vulnerability to malicious traffic without SDP protection. In contrast, the fourth scenario shows the network performance with SDP enabled under the same attack conditions. The throughput improved to an average value of 194.3 Mbits/s, illustrating SDP's effectiveness in mitigating the impact of DoS attacks and maintaining higher communication performance.

Figures 7 and 8 present the results of an IP scanning attack conducted using nmap [33] from UE 0, demonstrating the effectiveness of SDP in mitigating such attacks. After completing the 5G authentication and PDU session establishment through the UPF, UE 0 initiates an

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IP scanning attack. Upon PDU session establishment, UE 0 was assigned the IP address 10.45.0.2 and used the uesimtun0 interface in VM 1 to carry out the attack. Using nmap, UE 0 scanned subnet 192.168.56.0/24 through the UPF and detected seven active hosts. These hosts included critical components such as the military ZT manager (192.168.56.22), CN CP (192.168.56.110), UPF (192.168.56.112), and other devices (192.168.56.100, 104, 105, 111). These results occurred because all the components, including those constituting the core network, were placed within the same IP subnet, making them visible during the scanning process. However, this demonstrates the vulnerability of a network without an SDP, in which most devices can be easily enumerated.

After applying SDP, UE 0 performed the same IP scanning attack after completing 5G authentication and PDU session establishment. However, these results are significantly limited.

Using nmap, UE 0 scanned the same subnet (192.168.56.0/24) but was only able to detect two active hosts: the military ZT manager (192.168.56.22) and the UPF (192.168.56.112). Other critical components, including the CN, CP, application server, and additional devices, were no longer visible to attackers. This restriction occurred because the UPF, acting as the SDP AH, enforced policies to block unauthenticated devices, such as UE 0, from accessing unauthorized network components. The military ZT manager remained visible as it performed the SDP authentication and was explicitly permitted by the SDP AH policies to facilitate this process.

Table 3 and Figure 6 highlight the tradeoffs and benefits of using SDP in 5G networks. Although SDP introduces a slight overhead in non-attack scenarios, its ability to maintain a higher throughput and ensure secure communication under attack conditions underscores its critical role in guaranteeing the resilience and reliability of 5G networks in defense environments. Figures 7 and 8 show that the SDP system ensures that unauthorized devices cannot interact with or discover most network components, thereby enhancing the security and isolation of the defense networks.

```
root@VM5:/home/vm5# nmap -e uesimtun0 -sn 192.168.56.0/24
Starting Nmap 7.80 ( https://nmap.org )
Nmap scan report for 192.168.56.22
Host is up (0.0013s latency).
Nmap scan report for 192.168.56.112
Host is up (0.00090s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned in 17.65 seconds
```

FIGURE 8 IP scanning attack results when the SDP is applied to the network.

6 | EXPERIMENT RESULTS

In this section, we describe the experimental setup of the testbed, experimental scenario, and process of verifying the proposed architecture. Through this experiment, we demonstrate the feasibility and effectiveness of a ZT-architecture-based military network in a real 5G environment.

6.1 | Experiment setup and scenario

As shown in Table 4 and Figure 9, we implemented a testbed using a personal computer for the UE, three additional computers for the CN, a military ZT manager, and a web server, and an SDR for the gNB. For 5G communication between the UE and the CN via the SDR-based gNB, we used the OAI [19]. The CN was hosted on a PC

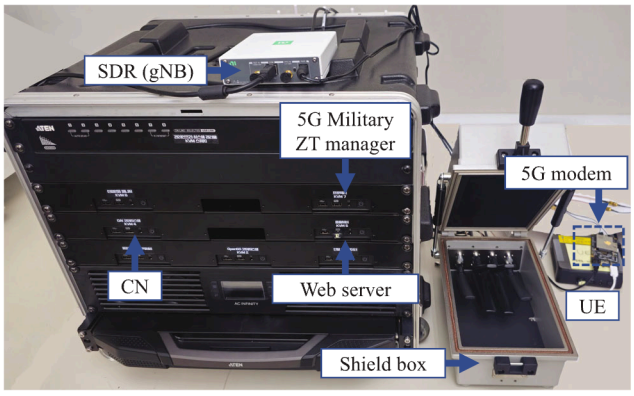


FIGURE 9 Real-world testbed of a ZT 5G defense network.

TABLE 4 Configuration details and assigned roles in experimental testbed for the zero-trust (ZT) defense network.

Device	SW and role	IP	Network	SDP role	OS
Mini-PC + 5G modem	5G UE	12.1.1.2	5G	SDP IH	Windows10
Software-defined radio (SDR)	OpenAirInterface RAN (gNB)	-		-	-
Mini-PC	OpenAirInterface CN	192.168.0.10		SDP AH	Ubuntu 22.04
	5G Military ZT manager	192.168.0.12	Defense	SDP controller	
	Server	192.168.0.14		SDP IH	

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running Ubuntu 22.04 LTS, while the gNB was implemented using a USRP B210 SDR [20], connected to the CN via USB. For the UE, we employed a Windows 10 mini-PC equipped with a Quectel RM500Q-GL module to provide 5G connectivity. This modem included a USIM registered with the CN. The experiment reused the SDP components described in Section 5, including the SDP IH program implemented in C++, the SDP AH software in Python, and the military ZT manager with a web GUI developed using the Spring Framework in Java.

In the 5G network, if the UE successfully connects to the CN, it is assigned an IP address, 12.1.1.2, by the CN's control functions. For the private military network, we assigned IP addresses 192.168.0.12 for the military ZT manager and 192.168.0.14 for the web server.

In this testbed scenario, the UE first established a 5G connection and then attempted to access the web server within a private defense network. The process began with a successful 5G connection between the UE and the CN, enabling communication over the 5G infrastructure.

The users then authenticate themselves through the SDP IH software by entering their unique ID and password and submitting the authentication request to the military ZT manager. Acting as the SDP controller, the ZT manager verifies the user's identity and authorizes the UE to access the network. To validate the SDP architecture, the system compared conditions before and after SDP authentication, confirming that an authenticated UE could successfully access the web server within the private network. Communication between the SDP, IH, AH, and controller forms the backbone of this process, determining access permissions. Additionally, the testbed implemented the SDP AH software, which modified the iptables within the UPF of the CN to enforce secure data routing and access control. These modifications ensured that only the authenticated UEs could interact with the private network. Screenshots of the SDP IH and military ZT manager interfaces further demonstrate the functionality of the system, highlighting the effectiveness of SDP in managing secure communication within the defense network.

6.2 | Experimental verification

Figure 10 shows the test results from the ZT 5G military network testbed, designed to validate the implementation of the ZT architecture within a defense 5G network. The green arrows represent the SDP control channel, which manages authentication and policy enforcement between the UE, CN, and military ZT manager. The yellow arrows indicate the SDP data channel, established following successful SDP authentication.

The blue arrows represent the 5G channels connecting the UE to the CN via the gNB, handling standard 5G communication processes. As shown in Figure 10 ①, the UE attempts to access the server within the defense network through 5G communication. Although the UE completes the 5G connection, it cannot access the server

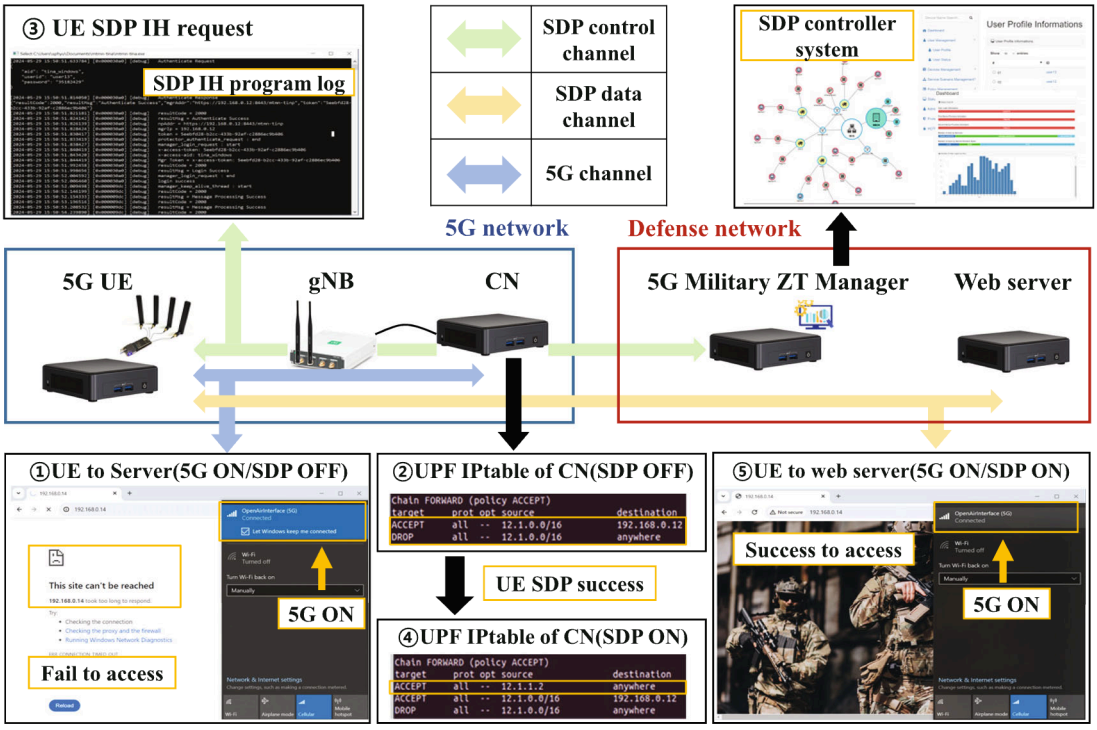


FIGURE 10 Performance and security validation results from the real-world testbed of the proposed ZT 5G defense network.

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(192.168.0.14) because SDP authentication has not yet succeeded. This demonstrates that, even with an active 5G link, the SDP blocks unauthorized access to the private network for security reasons.

Figure 10 ② shows the iptables firewall configuration of the UPF in the CN before the UE undergoes the SDP authentication process. The iptables configuration defines the IP addresses that are allowed or blocked from accessing the private network. In this default configuration, traffic with a source IP range of 12.1.0.0/16 and a destination IP address of 192.168.0.12 (military ZT manager) is permitted, while traffic with any other destination address is blocked. This restrictive policy is enforced when the SDP authentication is not active, ensuring that only predefined trusted UEs can communicate with the military ZT manager.

In Figure 10 ③, after establishing the 5G connection, the UE executes the SDP IH program and initiates the SDP authentication process with the military ZT manager. The log confirms a successful authentication, including the exchange of keep-alive messages to maintain the SDP control channel session. Once the UE is authenticated by the SDP controller, it is granted access to communicate with the defense network. After successful authentication, the military ZT manager sends the detailed information of the authenticated UE to the CN (SDP AH) via the SDP control channel.

Figure 10 ④ illustrates how the iptables rules in the UPF of the CN are updated after SDP authentication is activated. Once the military ZT manager authenticates the UE, it sends the details of the successfully authenticated UE to the SDP AH (CN) via the SDP control channel. In response, the CN adds an "anywhere" rule for the UE's IP address (12.1.1.2) in the UPF iptables. This "anywhere" rule allows the UE to access the server within the defense network.

In Figure 10 ⑤, the UE successfully accesses the server (IP: 192.168.0.14) after both 5G and SDP are active. With both the 5G network and SDP architecture enabled, the UE can securely communicate with the server in the defense network. The SDP ensures that the connection is protected, meeting the high security requirements of the ZT framework in a defense environment. Additionally, the GUI of the military ZT manager displays the 5G network topologies, user/device statuses, and any relevant communication statistics.

7 | DISCUSSION AND FUTURE WORK

The simulation results highlighted the advantages of incorporating SDP into 5G networks. The connection time results (Table 3) demonstrate that while SDP introduces a slight overhead owing to the authentication processes, it remains efficient for real-time applications in a 5G wireless environment. Additionally, the throughput results (Table 3 and Figure 6) emphasize SDP's resilience under DoS attack conditions, maintaining a significantly higher performance than non-SDP setups. The IP scanning attack simulations (Figures 7 and 8) further demonstrate SDP's ability to restrict unauthorized access to critical network components, effectively protecting the defense network from reconnaissance attacks. These findings emphasize SDP's capability to secure military networks against various threats while ensuring reliable performance.

The results of this experiment effectively demonstrate the process of verifying and securing a 5G military network using the ZT architecture. Based on these results, we verified how a combination of SDP and 5G networks can ensure that defense systems securely access critical resources while using a commercial 5G network infrastructure. The step-by-step breakdown of failed access attempts (when the SDP is off) and successful secure connections (when the SDP is on) illustrates the importance of ZT and SDP in securing military networks in 5G environments. Examples of failed access attempts (when the SDP is off) and successful secure connections (when the SDP is on) further demonstrate the importance of ZT and SDP in securing defense networks in 5G communication environments.

In the future, additional research will be conducted to improve the scalability and performance of ZT-based 5G defense network architectures. First, the 5G ZT testbed currently built uses commercial gNodeB and defense-only core networks to validate the proposed architecture. However, this system needs to be tested in a more complex, actual operational environment, as in [34]. Additionally, to analyze and optimize the performance of ZT-based 5G networks in various environments, research on their compatibility with gNB and other commercial 5G components is required. Second, it is necessary to introduce machine learning-based security models, such as those in [35, 36], to increase the scalability and security performance of the proposed SDP-based architecture. This approach enhances the efficiency of network anomaly detection and threat prediction.

8 | CONCLUSION

In this study, we explain how 5G technology is integrated into a defense network using both commercial and military-specific components based on a ZT architecture. Furthermore, this study demonstrates the successful implementation of the ZT architecture in a 5G network

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using a commercial gNB infrastructure. The simulation results further confirm the ability of the architecture to mitigate IP scanning and DoS attacks, ensuring secure and reliable communication under both normal and attack scenarios. The proposed architecture, verified using a real-world testbed, enhances security by integrating an SDP for secure access to private defense networks. The results highlight the practicality of leveraging commercial 5G elements to achieve extensive coverage and high security at a lower cost, making them viable solutions for future military communication needs.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

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